

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{e^{L\kappa_L}}{q(\kappa_L)} &\leq \frac{1}{f(\cos g_L(\kappa_L))} = \frac{1}{2 - \cos g_L(\kappa_L) - \sqrt{(2 - \cos g_L(\kappa_L))^2 - 1}} \\ &= 2 - \cos g_L(\kappa_L) + \sqrt{(2 - \cos g_L(\kappa_L))^2 - 1}. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \{X^1, X^3\} + \{X^4, X^2\} + \{X^5, X^7\} + \{X^8, X^6\} &= 0 \\ \{X^1, X^4\} + \{X^2, X^3\} + \{X^8, X^5\} + \{X^7, X^6\} &= 0 \\ \{X^1, X^5\} + \{X^4, X^8\} + \{X^7, X^3\} + \{X^6, X^2\} &= 0 \\ \{X^1, X^6\} + \{X^2, X^5\} + \{X^3, X^8\} + \{X^4, X^7\} &= 0 \\ \{X^1, X^7\} + \{X^3, X^5\} + \{X^8, X^2\} + \{X^6, X^4\} &= 0 \\ \{X^1, X^8\} + \{X^5, X^4\} + \{X^2, X^7\} + \{X^6, X^3\} &= 0. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ll} (P - SDP) : & (D - SDP) : \\ \text{maximize } b^T y & \text{minimize } C \cdot X \\ \text{subject to } C - \sum_{i=1}^m y_i A_i \succeq 0 & \text{subject to } A_i \cdot X = b_i \forall i \in \{1, \dots, m\} \\ & X \succeq 0, \end{array}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{lcl} \hat{g}_{\mu\nu} & = & g_{\mu\nu} + G_{mn} A^{(m)}_{\mu} A^{(n)}_{\nu}, \\ \hat{g}_{\mu m} & = & G_{mn} A^{(n)}_{\mu} = \hat{k}_{(m) \mu}, \\ \hat{g}_{mn} & = & G_{mn} = \hat{k}_{(m)}^{\hat{\mu}} \hat{k}_{(n) \hat{\mu}}. \end{array} \right.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\rho(\lambda) = \frac{4}{\pi} \left\{ \begin{array}{ll} \left(1 + \frac{2}{G}[1 - 2(1 - \lambda^2/2)^2]\right) / \sqrt{4 - \lambda^2} & 0 \leq \lambda \leq 2, \quad G \geq 2, \\ \frac{2\lambda}{G} \sqrt{(G/2 - (1 - \lambda^2/2)^2)} & \lambda_1 \leq \lambda \leq \lambda_2, \quad G \leq 2, \\ 0 & \text{otherwise,} \end{array} \right.$$